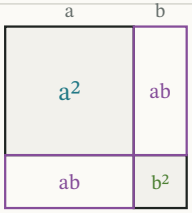
The binomial theorem CHEAT SHEET

MULTIPLYING OUT A BINOMIAL



the square of a + b:
a², two ab strips, b²

the cube of a + b: 8 blocks,
1 a³ · 3 a²b · 3 ab² · 1 b³

$(a+b)^2 = a^2 + 2ab + b^2$

$(a-b)^2 = a^2 - 2ab + b^2$ (not $a^2 + b^2$)

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

$(x+2)^3 = x^3 + 6x^2 + 12x + 8$

$99^2 = (100-1)^2 = 9801$ $103^2 = 10\,609$

$(10+0.1)^3 = 1000 + 30 + 0.3 + 0.001$

· $(a+b)^n$ has n + 1 terms
· powers of a fall by 1, powers of b rise by 1
· in every term the powers add up to n

WORKED EXPANSIONS

$(x-3)^2 = x^2 - 6x + 9$

$(2x-1)^3 = 8x^3 - 12x^2 + 6x - 1$

$(2x+3)^3 = 8x^3 + 36x^2 + 54x + 27$

$(3-x)^4 = 81 - 108x + 54x^2 - 12x^3 + x^4$

check any expansion: put x = 1 on both sides

PASCAL'S TRIANGLE

row 0 1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1
1 8 28 56 70 56 28 8 1
1 9 36 84 126 126 84 36 9 1
1 10 45 120 210 252 210 120 45 10 1

rows 0 to 10 · row n is the coefficients of (a + b) to the n

ITS PATTERNS

each number = the two above added: 3 + 3 = 6

each row is a mirror; row n adds up to 2^n

$11^2 = 121$ $11^3 = 1331$ $11^4 = 14\,641$

$11^5 = 161\,051$ (row 5 with carries)

4 coin flips, exactly 2 heads: 6 orders

$(x+2)^4 = x^4 + 8x^3 + 24x^2 + 32x + 16$

$(x-1)^5 = x^5 - 5x^4 + 10x^3 - 10x^2 + 5x - 1$

$1.01^6 = 1 + 0.06 + 0.0015 + 0.00002 + \ldots \approx 1.0615$

COUNTING: N! AND N CHOOSE R

multiply the choices: 4 shirts × 3 jeans = 12

$n! = n(n-1)(n-2)\ldots 2 \cdot 1$ $5! = 120$ $0! = 1$

$10! = 3\,628\,800$ podium of 3 from 7: $7 \cdot 6 \cdot 5 = 210$

$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$ said "n choose r" (also ${}^nC_r$)

$\binom{4}{2} = 6$ $\binom{5}{2} = 10$ $\binom{12}{5} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{5!} = 792$

$\binom{n}{0} = \binom{n}{n} = 1$ $\binom{n}{1} = n$ $\binom{n}{r} = \binom{n}{n-r}$

Pascal's rule: $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$

$\binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56 = \binom{7}{2} + \binom{7}{3} = 21 + 35$

exactly 7 right of 10: $\binom{10}{7} = \binom{10}{3} = 120$

ARRANGE OR CHOOSE?

order matters (arrange): $\frac{n!}{(n-r)!}$ 7 runners, 3 medals: 210

order doesn't (choose): $\binom{n}{r}$ 2 toppings of 4: 6

5 friends, every pair shakes hands: $\binom{5}{2} = 10$

WHY PASCAL'S RULE WORKS

a class of n + 1, one of them Sam; a team of r:
Sam in: choose the other r − 1 from n
Sam out: choose all r from the n
every team counted once: Pascal's rule

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THE BINOMIAL THEOREM

$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{n}b^n$

$(a+b)^n = \Sigma\binom{n}{r}a^{n-r}b^r, \quad r = 0, 1, \ldots, n$

why: choose b in r of the n brackets, $\binom{n}{r}$ ways

$(a+b)^7$: 1, 7, 21, 35, 35, 21, 7, 1

induction: $\binom{k}{r} + \binom{k}{r-1} = \binom{k+1}{r}$ (Pascal's rule)

SPECIAL CASES AND WORKED EXPANSIONS

$(x-y)^n$: the signs alternate

$(x-2y)^5 = x^5 - 10x^4y + 40x^3y^2 - 80x^2y^3 + 80xy^4 - 32y^5$

$(1+x)^n = 1 + nx + \binom{n}{2}x^2 + \ldots + x^n$

$\Sigma\binom{n}{r} = 2^n$ alternating sum $= 0$

$(x^2 + \frac{3}{x})^4 = x^8 + 12x^5 + 54x^2 + \frac{108}{x} + \frac{81}{x^4}$

$(1-2x)^5 = 1 - 10x + 40x^2 - 80x^3 + 80x^4 - 32x^5$

$(a+b)^4 - (a-b)^4 = 8a^3b + 8ab^3$

$(\sqrt{3}+\sqrt{2})^4 - (\sqrt{3}-\sqrt{2})^4 = 40\sqrt{6}$

THE GENERAL TERM

$T_{r+1} = \binom{n}{r}a^{n-r}b^r$

term k has $r = k-1$ (the 4th term: $r = 3$)

x^6y^3 in $(x+2y)^9$: $\binom{9}{3}\cdot 2^3 = 84 \cdot 8 = 672$

5th term of $(2x-1)^7$: $\binom{7}{4}(2x)^3(-1)^4 = 280x^3$

x^4 in $(3x-2)^6$: $r = 2$, $15 \cdot 81 \cdot 4 = 4860$

x^5 in $(2x-3)^8$: $56 \cdot 32 \cdot (-27) = -48\,384$

THE TERM INDEPENDENT OF X

independent of x: set the power of x to 0

$(x + \frac{1}{x})^6$: x^{6-2r}, $r = 3 \rightarrow 20$

$(x^2 + \frac{2}{x})^6$: x^{12-3r}, $r = 4 \rightarrow 240$

$(x - \frac{2}{x^2})^9$: $r = 3 \rightarrow 84 \cdot (-8) = -672$

$(x + \frac{1}{x})^5$: $r = 2.5$, so there is none

MIDDLE TERMS

n even: one middle term, term $\frac{n}{2} + 1$

n odd: two, terms $\frac{n+1}{2}$ and $\frac{n+3}{2}$

$(\frac{x}{3} + 9y)^{10}$: $T_6 = 61\,236x^5y^5$

$(x + \frac{1}{x})^7$: $T_4 = 35x$, $T_5 = \frac{35}{x}$

the middle coefficient is the largest: $\binom{12}{6} = 924$

APPROXIMATIONS

$(1+x)^n \approx 1 + nx$ when x is small

$1.01^{10} \approx 1 + 0.1 + 0.0045 + 0.00012 = 1.10462$

$1.01^{12} \approx 1 + 0.12 + 0.0066 + 0.00022 \approx 1.1268$

12 months at 1% ≈ 12.68%, not 12%

$0.99^5 \approx 1 - 0.05 + 0.001 = 0.951$ (0.95099)

$1.02^5 \approx 1.10408$: 50 000 → 55 204

BIG POWERS AND COMPARISONS

$98^5 = (100-2)^5 = 9\,039\,207\,968$

$101^4 = (100+1)^4 = 1\,04\,06\,04\,01$

$1.01^{1\,000\,000} > 1 + 1\,000\,000 \times 0.01 > 10\,000$

$1.1^{10\,000} > 1 + 10\,000 \times 0.1 = 1001 > 1000$

REMAINDERS

$6^n = (1+5)^n = 1 + 5n + 25k$

so $6^n - 5n$ leaves remainder 1 on ÷ 25

$9^{n+1} - 8n - 9 = 64k$ $\quad 7^n - 6n - 1 = 36k$

$11^{10} = (1+10)^{10}$ ends in 01